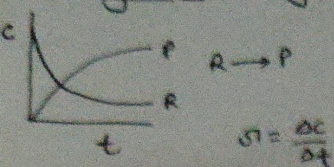


chemical kinetics

Rate of rxn / Velocity / speed of rxn



Unit of $r = M \text{ time}^{-1}$

$= \text{mole L}^{-1} \text{sec}^{-1}$

Type of Rate (or)

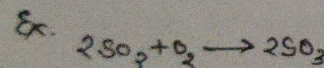
Avg. (or) Inst. (or)

$$r_{\text{avg}} = \frac{C_2 - C_1}{t_2 - t_1} \quad r_{\text{inst}} = \frac{dc}{dt} = -\frac{dR}{dt} = -\frac{dP}{dt}$$

Representation of ROR in terms of concⁿ of Reactant or product:



$$r = -\frac{1}{n_1} \frac{dA}{dt} = -\frac{1}{n_2} \frac{dB}{dt} = \frac{1}{m_1} \frac{dC}{dt} = \frac{1}{m_2} \frac{dD}{dt}$$



$-\frac{dSO_2}{dt}$ = rate of disappearance of SO_2

$\frac{dSO_3}{dt}$ = rate of appearance (formation) of SO_3



rate of disapp. of O_2 is

$$1.6 \times 10^{-2} \text{ kg L}^{-1} \text{ min}^{-1}$$

rate of formation of SO_3

$$\text{kg L}^{-1} \text{ min}^{-1}$$

$$-\frac{1}{2} \frac{d(SO_2)}{dt} = -\frac{d(O_2)}{dt} = \frac{1}{2} \frac{dSO_3}{dt}$$

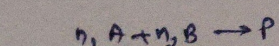
$$\left(\frac{1}{32} \right) \times 1.6 \times 10^{-2} = \frac{1}{2} \left(\frac{1}{80} \right) \frac{dSO_3}{dt}$$

$$\frac{dSO_3}{dt} = 8 \times 10^{-2}$$

Rate law

In any chemical rxn $ROR \propto \text{conc}^n \text{ of Reactant}$ which is experimentally Proved

* Rate law is Experimental concept



$$r = K(A)^{n_1}(B)^{n_2}$$

Order of Rxn (O.R)

Total sum of power of concⁿ in Rate law k/a O.R

* OR Experimental concept

OR may be +ve, -ve, x Neutral, whole no. or fraction.

$$r = K(A)^p(B)^q$$

$$O.R = p + q$$



mech.



$$r = K(O_3)(O) \quad K_c = \frac{(O_2)(O)}{O_3} \quad (O) = K_c(O_3)(O_2)$$

$$r = K(O_3) K_c \frac{O_3}{O_2}$$

$$r = K(O_3)^2(O_2)^{-1}$$

$$r = 2 - 1 = 1$$

Effects of concⁿ on rate of rxn:

$$(1) r = K(A)^n$$

$$\frac{r_2}{r_1} = \left(\frac{A_2}{A_1} \right)^n$$

$$(2) r = K(A)^p(B)^q$$

$$\frac{r_2}{r_1} = \left(\frac{A_2}{A_1} \right)^p \left(\frac{B_2}{B_1} \right)^q$$

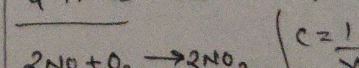
$$(3) PV = nRT$$

$$P = CRT$$

$$C = \frac{n}{V}$$

$$C \propto \frac{1}{V}$$

Ex. Given gaseous rxn is an elementary rxn if vol^m of container increase of 4 times new rate?



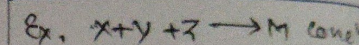
$$r = K(NO)^2(O_2)^1$$

$$\left(\frac{1}{4} \right)^2 \left(\frac{1}{4} \right) = \frac{1}{64}$$

change in concⁿ

change in ROR

2 time	4 times
2 time	2 times
2 time	No change
2 time	8 times
2 time	1/2 times
2 time	1/4 times
2 time	1.414 times
2 time	2.82 times
3 time	27 times
3 time	1/9 times
8 time	1.732 times
3 time	5.79 times
8 times	2 times



of only 'X' rises 27 time Rate increases 9 times rate of rxn does NOT changes with change in concⁿ of Y.

When concⁿ of Z rises to 8 times rate will double Rate law = ?

$$r = K(X)^1(Y)^2(Z)^n$$

$$\frac{r_2}{r_1} = \left(\frac{X_2}{X_1} \right)^1 \left(\frac{Y_2}{Y_1} \right)^2 \left(\frac{Z_2}{Z_1} \right)^n$$

$$9 = (27)^n$$

$$3^2 = 3^{3n}$$

$$P = \frac{2}{3}$$

$$Q = 0$$

$$2 = (8)^n$$

$$2^1 = 2^{3n}$$

$$\frac{1}{3} = n$$